

Survival After Proton-Beam Irradiation of Uveal Melanomas

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• **PURPOSE:** To evaluate the independent prognostic factors for survival, metastasis, local recurrence, and enucleation in patients who had undergone proton-beam therapy for posterior uveal melanomas.

• **DESIGN:** Interventional case series.

• **METHODS:** In this retrospective study, 224 consecutive incident cases were treated at the Biomedical Cyclotron Centre (Nice, France) from June 1991 to December 1997. Overall, metastasis-free, local recurrence-free, and enucleation-free survival rates were calculated according to the Kaplan-Meier method using the log-rank test. The multivariate prognostic analysis was performed using the Cox proportional hazards model.

• **RESULTS:** The 5-year overall survival rate was 78.1% (SE: 3.7%). A largest basal tumor diameter (LTD) below 10 mm and female sex were independently associated with a better prognosis. The 5-year metastasis-free survival rate was 75.6% (SE: 3.6%). Only an LTD above 10 mm and ciliary body involvement were independently associated with metastasis. Ten patients (4.5%) had a local recurrence, which was correlated with the risk of metastasis ($P = .045$). The 5-year enucleation-free survival rate was 69.6% (SE: 4.0%). Once again, an LTD below 10 mm and female sex were predictive of a better prognosis.

• **CONCLUSION:** Our results with proton-beam therapy correspond to those reported in the literature. This treatment strategy is safe and yields predictably good results. In

addition to the two independent prognostic factors for survival and metastasis, namely LTD and ciliary body involvement, sex also had a significant impact in our case series, but the clinical relevance of this finding is unknown. (Am J Ophthalmol 2004;137:1002–1010. © 2004 by Elsevier Inc. All rights reserved.)

MELANOMAS OF THE UVEA ARE THE MOST COMMONLY occurring type of primary intraocular tumor in adults,¹ and may arise in either the anterior (that is, iridial) or the posterior (that is, ciliary body or choroidal or both) portions of this tract. Conservative treatment of uveal melanomas includes brachytherapy using different sources of radiation, transscleral local resection, transpupillary thermotherapy, argon-laser photocoagulation, and external-particle radiotherapy.² Proton-beam therapy was first instigated for the handling of uveal melanomas in 1975 by Gragoudas and associates.³ Several retrospective^{4–9} and prospective studies conducted by the Collaborative Ocular Melanoma Study (COMS) Group^{10,11} have yielded no evidence in favor of any particular therapeutic strategy (that is, radiation or enucleation). However, external-beam irradiation with accelerated proton particles offers several theoretical advantages over plaque therapy, including an optimal and uniform delivery of radiation to the entire tumor, minimal radiation damage to the surrounding normal tissue and no handling of radioactive elements by the ophthalmologist. We have used this mode of treating uveal melanomas since 1991, when it first became available in our region.

In the present retrospective, interventional case series study, we wished to determine the independent prognostic factors for overall survival, metastasis-free survival, local recurrence-free survival and enucleation-free survival in patients treated for posterior uveal melanomas by proton-beam therapy.

METHODS

• **PATIENTS AND TUMORS:** Two hundred twenty-four consecutive patients with posterior uveal melanomas were

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